

Recitation: Factoring, Roots, & Discriminants

Problem 1. If any of the following polynomials factor, factor them completely:

(a) $x^2 - 10x - 24$

(f) $x^2 + 1$

(b) $25x^3 - 4x$

(g) $x^4 + x^3 - 2x^2$

(c) $x^2 - 7x - 6$

(h) $3x^2 - x - 2$

(d) $x^3 + 8$

(i) $x^3 - 2x^2 + 5x - 10$

(e) $4x^2 + 8x + 3$

(j) $32x^2 - 76x + 35$ [Hint: This has roots $\frac{5}{8}, \frac{7}{4}$.]

Problem 2. If $p(x)$ is a polynomial has roots $x = -1, -1, -1, 1, 3$ and y -intercept 5, determine the polynomial. Also, determine the domain and range of $p(x)$.

Problem 3. Find all possible solutions to the following:

$$\frac{5}{2-x} = x$$

Problem 4. Explain why the polynomial $100x^2 - 400x + 100$ does not factor ‘nicely’. Then use the quadratic formula to factor the polynomial completely.